# Fenrix Strategy Validation Lab

Backtesting shows where a strategy worked. Fenrix searches for where it breaks.

- Product flow: Describe → Review → Lock → Backtest → Stress → Minimize → Export
- Strategy hash (immutable): c9d19b2e8d20667a…
- git SHA: d6060783173d94586f6ceac10beda05bd34d4d31 · data mode of record: yfinance · universe: 30 assets

# The problem

## A green backtest is a claim, not a validation

• Backtests only tell you where a strategy already worked — they are silent about fragility.
• Students and retail quants ship strategies validated on one historical path.
• Overfitting, cost blindness and regime fragility hide until real capital finds them.
• There is no cheap, reproducible way to ask: under which conditions does this break?

# Product workflow

Describe → Review → Lock → Backtest → Stress → Minimize → Export

- Describe: plain-English strategy → structured clause ledger (every clause reviewed).
- Review + Lock: mandatory approve step → immutable version + canonical SHA-256 hash.
- Backtest: the SAME locked hash runs a real multi-asset historical backtest.
- Stress: the SAME hash enters a sealed synthetic failure search across mechanisms.
- Minimize: confirmed failures are shrunk to a minimal reproducible counterexample.
- Export: evidence package with manifest, hashes and claim ledger.

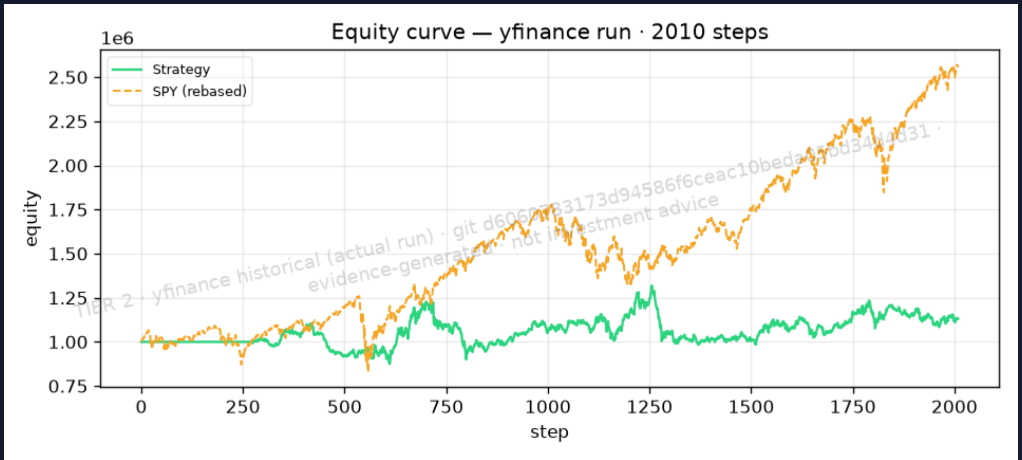
# Demo run — Real historical backtest (yfinance)

30 assets · 50 trades · all figures from deck_data.json

| 13.07% | 0.18 | -26.75% | 14.65% | 14.03% | 50 |
|---|---|---|---|---|---|
| cumulative return | Sharpe | max drawdown | ann. volatility | cost % of capital | trades |

- CAGR 1.55% · Sortino 0.22 · Calmar 0.06
- Benchmark CAGR 12.49% · information ratio -0.46
- Gross exposure avg 103.70% · net exposure avg 0.00%
- Costs are explicit bounded heuristics (commission/spread/slippage/borrow) — not broker calibrations.

TIER 2



# What the normal backtest missed

## Baseline vs confirmed stress failure

- Baseline (Real historical backtest (yfinance)): Sharpe 0.18, max drawdown -26.75%, cumulative return 13.07%.
- Sealed stress search: 8 confirmed failures out of 24 worlds (33.33% failure rate).
- Confirmed-failure mechanisms: borrow_cost_increase, correlation_breakdown, slippage_inflation, volatility_expansion.

TIER 2

# Minimized counterexample

Smallest reproducible world where the locked strategy still fails

- Mechanism: volatility_expansion · seed 20264.
- Intensity minimized 1.038 → 0.9285 and it STILL fails: True.
- Violated predicates: n/a.
- Adjacent pass: no adjacent pass within radius — reported honestly, not fabricated.

TIER 2

# Education & research value

## A falsification lab, not a profit promise

- Students see WHY a strategy fails — mechanism, intensity, seed — not just an equity curve.
- Every claim is bound to a hash and an evidence file; grading and review become reproducible.
- Instructors can replay the minimized counterexample deterministically.
- Mechanism library searched this run: borrow_cost_increase, correlation_breakdown, delayed_rebalance, missing_data_shock, momentum_reversal, short_unavailability, slippage_inflation, spread_inflation, universe_churn, volatility_compression, volatility_expansion.

# Architecture & moat

## Immutable contracts + sealed synthetic worlds

• Clause ledger → canonical hash: the same immutable artifact flows through every stage.
• Sealed synthetic world generator: seeds + mechanisms are reproducible and tamper-evident.
• Evidence packages carry SHA-256 manifests binding data, backtest, campaign and replay.
• Failure minimizer + adjacent-pass search turn red flags into actionable counterexamples.
• Moat: the corpus of confirmed, minimized failures compounds with every run.

# Honest boundary & roadmap

## What this is NOT — and what comes next

- Synthetic stress worlds are generated, not historical; they probe fragility, not real future risk.
- Costs are explicit bounded heuristics (flat bps), not validated broker calibrations.
- Fenrix fundamentals (if used) are not point-in-time; lagged approximation only.
- No claim of live profitability, production execution fidelity, or universal realism.
- Failure search is not exhaustive; it reports confirmed failures only.
- Roadmap: point-in-time fundamentals, broker-calibrated costs, richer mechanism library, live paper-trading bridge.

# Ask

## Help us make strategy falsification the default

- Pilot with quant-finance classrooms and student funds this semester.
- Feedback on mechanism realism from practitioners.
- Compute credits for larger sealed stress campaigns.
- Reproduce everything: make submission-demo · make verify-submission · make pitch-deck.